MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548359

(CONVERTER, RANGE D/A, 4A16)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification, together with the general requirements of MIL-C-50652, comprise the complete requirements for the range D/A converter used in Computer, Range CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U. S. Army, Frankford Arsenal

F10548359	Range D/A Converter

PACKAGING DATA SHEET

10548359	Packaging of Circuit Card Assembly, Range D/A Converter

(Copies of specifications, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548359 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the circuit card assembly.

3.5 Performance. - For the requirement of 3.5.4 through 3.5.6 logic input levels shall be as follows:

Logic 0: 0.0 ±0.5 volt direct current (Vdc)
Logic 1: +3.5 ±1.0 Vdc level

3.5.1 Input voltages. - The input voltages shall be as specified in Table I. All voltages shall be regulated with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

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TABLE I. - INPUT VOLTAGES

CHARACTERISTIC	TERMINALS
-15.0 ±0.1 Vdc	P1-20, P1-21
-15.0 V return	P1-22, P1-23
+5.0 ±0.3 Vdc	P1-30, P1-31, P1-32
+5.0 V return	P1-24, P1-25, P1-26
+25.0 ±0.1 Vdc	P1-55, P1-56
+25.0 V return	P1-53, P1-54
-25.0 ±0.1 Vdc	P1-49, P1-50
-25.0 V return	P1-51, P1-52

3.5.2 Bias voltage. - The bias voltage measured at TP4 of the range D/A converter assembly shall be between the limts of +2.0 and +2.2 Vdc.

3.5.3 20-volt reference. - The reference voltage measured at P1-39 shall be between the limits of +19.90 and +20.10 Vdc.

3.5.4 5000 meter adjust. - With the logic levels applied as specified in step 1 of Table II, the output voltage at P1-41 shall be adjustable by R4 between the limits as specified in step 1 of Table II.

3.5.5 Range voltages.

3.5.5.1 Incremental limits. - With the logic levels applied sequentially for steps 2 through 12 as specified in Table II the range output voltages at P1-41 shall be within the limits as specified in Table II.

3.5.5.2 Cumulative limits. - With the logic levels applied sequentially as specified in Table III, the range output voltages at P1-41 shall be within the limits as specified in Table III.

3.5.6 Range ratios. - With the logic levels applied sequentially as specified in Table IV, the ratio of the range output voltages at P1-41 to the reference output voltage at P1-39 shall be within the limits as specified in Table IV.

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TABLE II. - RANGE VOLTAGES INCREMENTAL LIMITS

STEP	EQUIVALENT RANGE (M)	INPUT LOGIC LEVELS - TERMINALS P1										OUTPUT VOLTAGE P1-41	
		19	11	18	17	2	4	15	5	6	33	NOT LESS THAN	NOT MORE THAN
1	5000	0	0	0	1	0	1	1	1	1	1	+19.950	+20.050
2	2560	0	0	0	0	0	0	0	0	0	1	+10.140	+10.340
3	1280	0	0	0	0	0	0	0	0	1	0	+ 5.070	+ 5.170
4	640	0	0	0	0	0	0	0	1	0	0	+ 2.520	+ 2.600
5	320	0	0	0	0	0	0	1	0	0	0	+ 1.240	+ 1.320
6	160	0	0	0	0	0	1	0	0	0	0	+ 0.600	+ 0.680
7	80	0	0	0	0	1	0	0	0	0	0	+ 0.280	+ 0.360
8	40	0	0	0	1	0	0	0	0	0	0	+ 0.120	+ 0.200
9	20	0	0	1	0	0	0	0	0	0	0	+ 0.040	+ 0.120
10	10	0	1	0	0	0	0	0	0	0	0	+ 0.020	+ 0.080
11	5	1	0	0	0	0	0	0	0	0	0	+ 0.000	+ 0.060
12	0	0	0	0	0	0	0	0	0	0	0	- 0.030	+ 0.020

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TABLE III. - RANGE VOLTAGES CUMULATIVE LIMITS

STEP	EQUIVALENT RANGE (M)	INPUT LOGIC LEVELS TERMINALS P1- 19	11	18	19	12	4	15	5	6	33	OUTPUT VOLTAGE P1-41 NOT LESS THAN	NOT MORE THAN
1	5115	1	1	0	0	0	0	0	0	0	0	+20.255	+20.665
2	2555	1	1	1	0	0	0	0	0	0	0	+10.118	+10.322
3	1275	1	1	1	1	0	0	0	0	0	0	+ 5.049	+ 5.151
4	635	1	1	1	1	1	0	0	0	0	0	+ 2.500	+ 2.580
5	315	1	1	1	1	1	1	0	0	0	0	+ 1.220	+ 1.300
6	155	1	1	1	1	1	1	1	0	0	0	+ 0.580	+ 0.660
7	75	1	1	1	1	1	1	1	1	0	0	+ 0.260	+ 0.340
8	35	1	1	1	1	1	1	1	1	1	0	+ 0.100	+ 0.180
9	15	1	1	1	1	1	1	1	1	1	1	- 0.020	+ 0.100

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TABLE IV. - RANGE RATIO LIMITS

STEP	EQUIVALENT RANGE (M)	INPUT LOGIC LEVELS TERMINALS P1- 19	11	18	17	2	4	15	5	6	33	OUTPUT RATIO VOLTAGES NOT LESS THAN	NOT MORE THAN
1	5000	0	0	0	1	0	1	1	1	1	1	+0.9925	+1.0075
2	2560	0	0	0	0	0	0	0	0	0	1	+0.5044	+0.5196
3	1280	0	0	0	0	0	0	0	0	1	0	+0.2522	+0.2698
4	640	0	0	0	0	0	0	0	1	0	0	+0.1254	+0.1307
5	320	0	0	0	0	0	0	1	0	0	0	+0.0617	+0.0663
6	160	0	0	0	0	0	1	0	0	0	0	+0.0299	+0.0342
7	80	0	0	0	0	1	0	0	0	0	0	+0.0139	+0.0809
8	40	0	0	0	1	0	0	0	0	0	0	+0.0060	+0.0100
9	20	0	0	1	0	0	0	0	0	0	0	+0.0020	+0.0060
10	10	0	1	0	0	0	0	0	0	0	0	+0.0010	+0.0040
11	5	1	0	0	0	0	0	0	0	0	0	+0.0000	+0.0030
12	0	0	0	0	0	0	0	0	0	0	0	-0.0015	+0.0010

MIL-C-50652/14 (MI)

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4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table V shall be utilized in the performance of the applicable tests as specified in 4.6.3.

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TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power Supply	5.0 ±0.3 Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity 200 mA.
2. Power Supply	15.0 ±0.3 Vdc, regulated, ac ripple 50 mV peak-to-peak maximum; capacity 200 mA.
3. Power Supply	25.0 ±0.3 Vdc, regulated; ac ripple 50 mV peak-to-peak maximum; capacity 200 mA.
4. Power Supply	Same as item 3.
5. Power Supply	10.0 ±0.01 Vdc, regulated; ac ripple 5 mV peak-to-peak maximum; capacity 100 mA.
6. Digital volt-ratio meter	Visual display: 5 digits. Dc range: 0.0000 to ±999.99 V. Accuracy, ±0.01% of reading +0.001% of full scale dc input resistance greater than 10 megohms. Dc ratio 1.0999 to 1; accuracy ±0.0005% of reading +0.002% of full scale with ±20 V minimum reference. Common mode rejection to dc, greater than 134 dB; to 60 Hz, greater than 120 dB. Response time: 1500 ns maximum.
7. Test Fixture	As specified in Figure 1. See Table VI for test fixture parts list.

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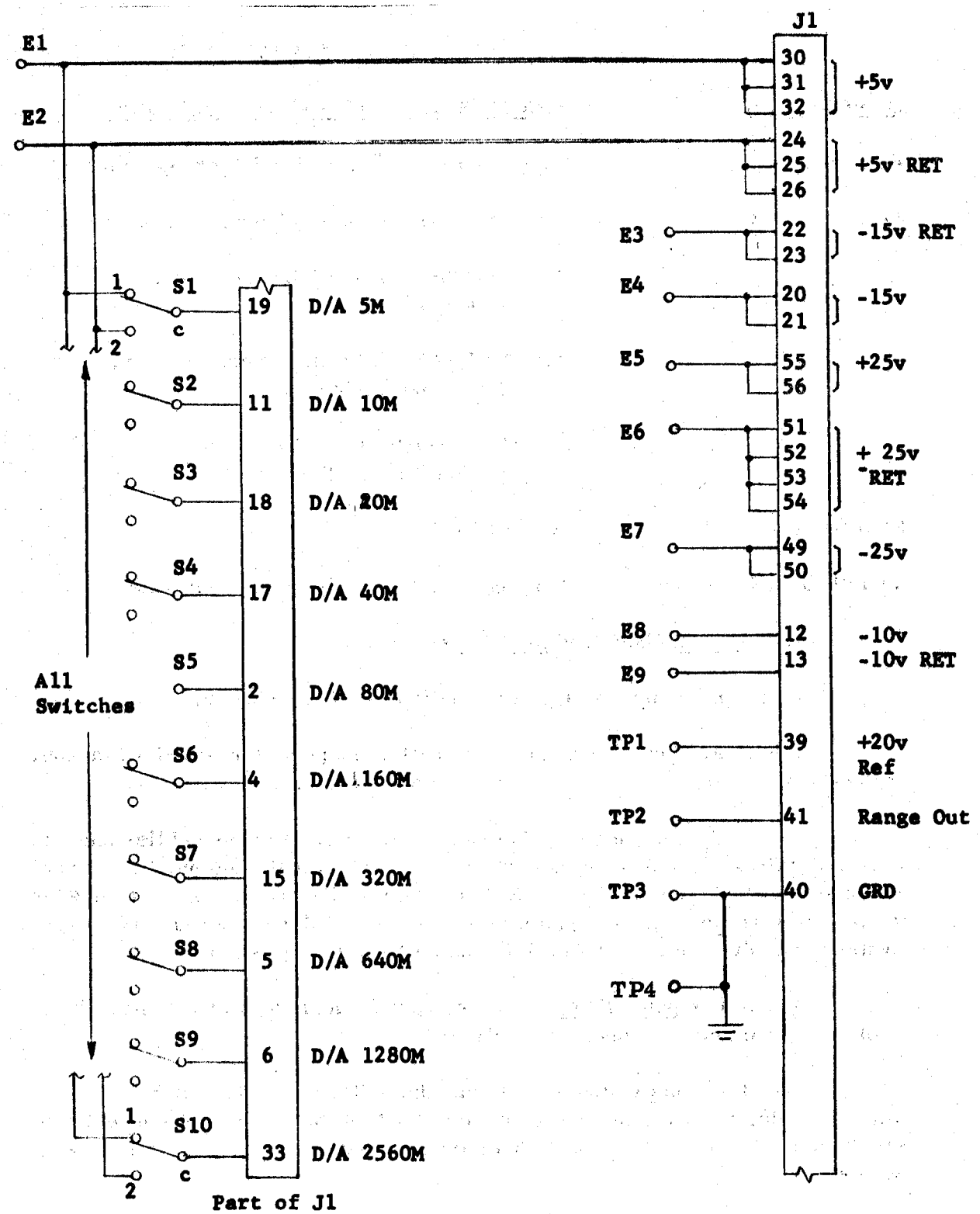


FIGURE 1. TEST FIXTURE SETUP

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TABLE VI. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1 thru E9	Binding post: H. H. Smith 257 or equivalent.
J1	Connector: any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco. Corp., Willow Grove, Pa.
S1 thru S10	Switch SPDT
TP1 thru TP4	Tip jack: H. H. Smith 1501 or equivalent.

4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-59652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The range D/A converter assemblies shall be tested as follows: Connect range D/A converter and test equipment as shown in Figure 2. Switches S1 through S10, on the test fixture, item 7, are used to supply the logic level input signals as specified in Tables II through IV. These switches supply a logic 1 in position 1 and a logic 0 in position 2.

4.6.3.1 Input voltages. - Using power supplies, items 1 through 5, apply input voltages as specified in Table I.

4.6.3.2 Bias voltages. - Using the volt-ratio meter, item 6, in the voltage mode, measure the bias voltage at TP4 of the range D/A converter to be as specified in 3.5.2. Use TP3 on the test fixture, item 7, as a ground reference.

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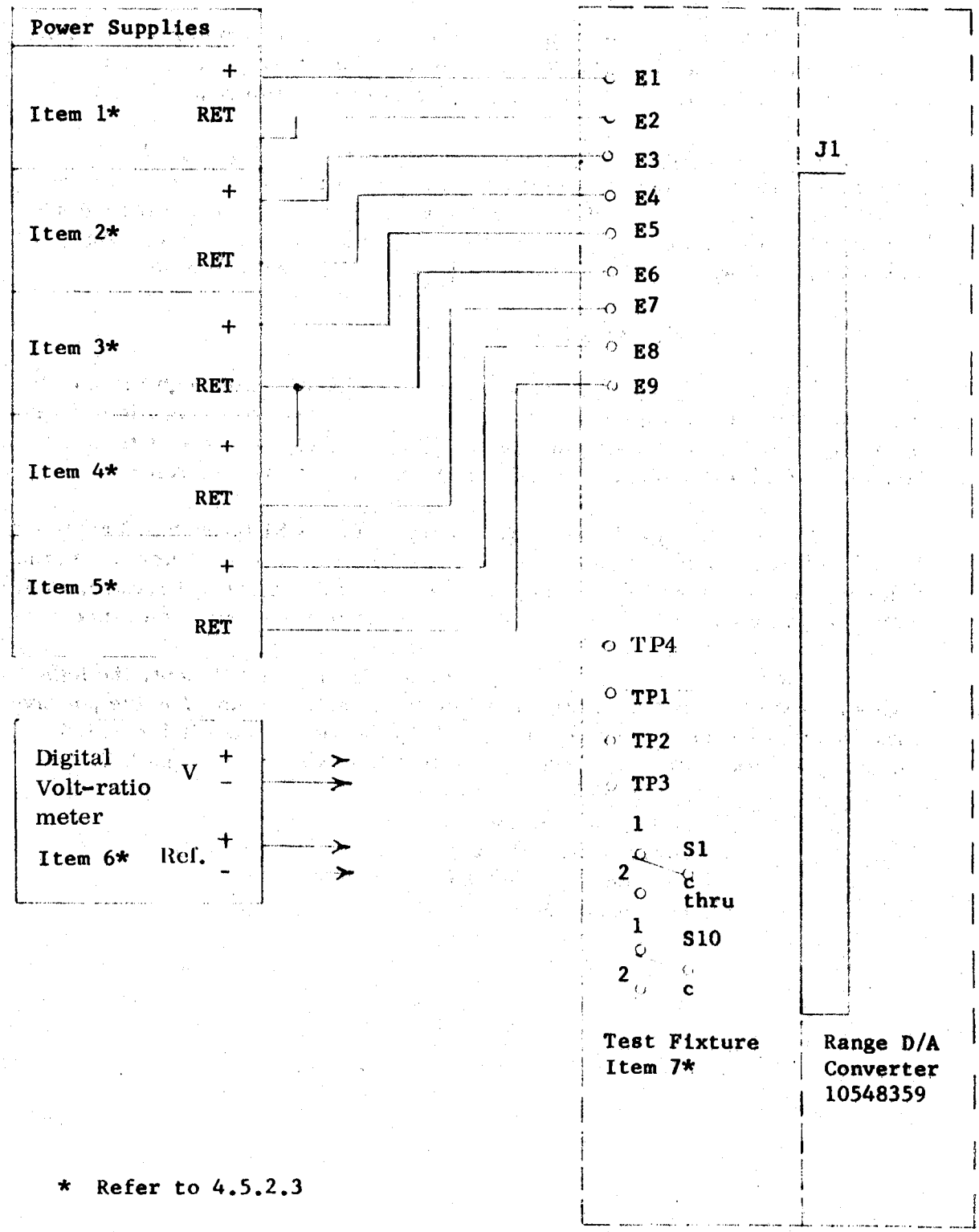


FIGURE 2. RANGE D/A CONVERTER TEST SETUP

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4.6.3.3 20-volt reference. - Using the volt-ratio meter, item 6, in the voltage mode, measure the 20-volt reference at TP1 to be as specified in 3.5.3. Use TP3 on the test fixture, item 7, as ground reference.

4.6.3.4 5000 meter adjust. - Using switches S1 through S10 on the test fixture, item 7, apply the logic levels as specified in step 1 of Table II. Using the volt-ratio meter, item 6, in the voltage mode, measure the voltage at TP2 on the test fixture to be adjustable, with R4, between the limits as specified in 3.5.4. Use TP3 on the test fixture as ground reference.

4.6.3.5 Range voltages.

4.6.3.5.1 Incremental test. - Using switches S1 through S10 apply the logic levels as specified in 3.5.5.1. Use the volt-ratio meter, item 6, in the voltage mode, to measure the range voltages at TP2 on the test fixture to be as specified in 3.5.5.1. Use TP3 on the test as ground reference.

4.6.3.5.2 Cumulative test. - Using switches S1 through S10 apply the logic levels as specified in 3.5.5.2. Use the volt-ratio meter, item 6, in the voltage mode, to measure the range voltages at TP2, on the test fixture, to be as specified in 3.5.5.2. Use TP3 on the test fixture as ground reference.

4.6.3.6 Ratio test. - Using switches S1 through S10 apply the logic levels as specified in 3.5.6. Use the volt-ratio meter, item 6, in the positive ratio mode to measure the range ratio voltages to be as specified in 3.5.6. Connect the ratio-volt meter leads to the test fixture test points as follows:

\+ voltage lead - TP2
\- voltage lead - TP3
\+ reference lead - TP1
\- reference lead - TP4

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10548359. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The range D/A converter is used in Computer, Range CP-888/VPS-2, part of Radar Set AN/VPS-2. The range D/A converter receives the range word in binary form and converts it to an analog voltage representing range. The converter also furnishes a +20 Vdc reference voltage.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

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Preparing Activity:

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Project No. 1285-A251

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